

# Leveraging Transformer Models for Extractive Question Answering in Criminal Report Analysis

Lenin G. Falconi<sup>1</sup>[0000–0003–4402–6643],  
Lorena Isabel Barona-López<sup>1</sup>[0000–0002–5184–3759], and  
Myriam Hernandez-Alvarez<sup>1</sup>[0000–0003–4718–0400]

Escuela Politécnica Nacional, Quito 170525, Ecuador

<https://www.epn.edu.ec>

{lenin.falconi, lorena.barona, myriam.hernandez}@epn.edu.ec

**Abstract.** Transformer-based models have significantly advanced Extractive Question Answering (EQA) in Natural Language Processing (NLP), yet their adaptation to legal document analysis remains understudied. This paper presents a domain-adapted BERT model for EQA on Spanish-language robbery case reports from Ecuador’s Prosecutor’s Office (*Fiscalía General del Estado*). Our methodology employed Generative Artificial Intelligence (GAI) to synthesize a dataset specifically tailored to robbery case reports in Stanford Question Answering Dataset (SQuAD)-format. We first evaluated the baseline performance of a pre-trained BERT model on this domain-specific dataset before conducting fine-tuning to adapt it to the robbery reports. The fine-tuned model demonstrated superior performance in processing criminal case information, achieving a final F1-score of 82.88. These results highlight the effectiveness of fine-tuning transformer-based models for EQA tasks in legal language processing and emphasize their potential to enhance information retrieval within judicial systems. Such advancements could improve service delivery to communities by optimizing workloads.

**Keywords:** Legal Natural Language Processing · NLP · Natural Language Processing · transformers · LNLN · law.

## 1 Introduction

Transformer-based models have emerged as powerful tools in NLP, enabling more accurate and efficient extraction of relevant information from complex texts such as criminal reports. These models utilize self-attention mechanisms to weigh the importance of different words in a sentence, allowing for better context understanding and improved performance in tasks like Question Answering (QA). By fine-tuning these models on domain-specific datasets, researchers can enhance their ability to discern pertinent details and provide precise answers to inquiries related to criminal incidents.

The implementation of EQA systems in legal contexts not only enhances the efficiency of information retrieval but also addresses challenges related to

data variability and complexity inherent in criminal reports. As these models are fine-tuned on diverse datasets, they can improve their generalizability across different types of legal documents, ensuring that critical details are accurately captured regardless of nuances in language or structure [8]. Moreover, recent advancements in model architectures, such as the integration of dynamic masking techniques, have shown promise in further refining performance by enabling models to better adapt to varying levels of difficulty in questions posed about complex narratives [13]. This evolution highlights the potential for transformer-based models to facilitate more informed decision-making processes within the justice system, ultimately contributing to enhanced outcomes for legal practitioners and stakeholders alike.

The Prosecutor’s Office of Ecuador is responsible for handling a wide range of cases, including criminal investigations and public safety matters. Among these, robbery poses a significant challenge, with 67,705 reported incidents in 2024<sup>1</sup> (equivalent to 376 robberies per 100,000 inhabitants). The comprehensive analysis of these crime reports serves multiple critical functions: (1) it enables more effective investigations by revealing criminal patterns, (2) supports victims’ insurance claims for stolen property, and (3) provides essential data for public safety planning. Artificial Intelligence (AI) is particularly valuable in this context, as automated systems can enhance the efficiency of processing case information, alleviate backlogs, and ultimately improve service delivery to citizens.

Our previous research investigated robbery classification using state-of-the-art neural architectures, with a focus on fine-tuned language models. Extending this work, the current study investigates the application of EQA systems for processing criminal reports. The proposed methodology demonstrates significant potential to assist legal authorities by: (1) extracting actionable insights for criminal investigations, and (2) enabling comprehensive crime statistics analysis. To our knowledge, this represents the first application of transformer-based models utilizing a specially curated Spanish-language SQuAD dataset derived from Ecuadorian legal documents.

This paper is organized as follows: Section 2 surveys existing applications of AI in legal domains, with a focus on question answering (QA) tasks. Section 3 describes our methodology, including dataset construction and model training. Section 4 presents the evaluation of our fine-tuned model. Finally, Section 5 summarizes our main findings and outlines opportunities for future work.

## 2 Literature Review

Large Language Models (LLMs) represent a category of Deep Learning (DL) models that integrate transformer architectures with large-scale pre-training on extensive text corpora. Trained in a self-supervised fashion, these models capture detailed linguistic patterns and can subsequently be fine-tuned to address

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<sup>1</sup> Statistics reported by FGE with data cut-off date June 13, 2025

domain-specific challenges in Legal NLP (LNLP). For instance, one influential model is described in [4], and another is presented in [15], where OpenAI reported that the model achieved performance at the 90th percentile on the Uniform Bar Exam, a standardized examination used to evaluate the competency of prospective attorneys. However, subsequent analysis by [11] indicates that GPT-4’s actual performance ranges from the 48th to the 62nd percentile. These authors critique the imprecision and lack of transparency in OpenAI’s report, arguing that such shortcomings may hinder the development of safe AI and lead to misconceptions regarding the model’s true capabilities in handling complex legal tasks.

Legal Question Answering (LQA) is a complex task that entails addressing queries related to legal issues, necessitating not only the analysis of extensive legal corpora but also their nuanced interpretation [2]. A wide range of models and algorithms have been explored for QA in NLP, such as *Recurrent Neural Networks (RNNs)*, *Long Short-Term Memory (LSTM)*, and *Convolutional Neural Networks (CNN)*. Additionally, information retrieval approaches, including *Term Frequency-Inverse Document Frequency (TF-IDF)*, are frequently utilized. Both extractive and generative methodologies have demonstrated effectiveness in addressing NLP-based QA tasks [10].

It is important to clarify the primary distinction between these two approaches. EQA identifies and retrieves a span of text directly from the given context: Given a tuple composed of the question  $q$  and the context  $c$ , the model predicts the start and end positions of the answer. In contrast, Generative Question Answering (GQA) synthesizes responses using encoder-decoder architectures, generating answers in an autoregressive manner [10]. According to a systematic evaluation of QA models by [10], generative readers generally outperform extractive readers with longer contexts, providing more fluent and comprehensive answers, whereas extractive readers are more effective when the context is limited and exhibit better out-of-domain generalization.

## 2.1 Large Language Models and Generative AI

GAI refers to a class of AI models that learn the underlying data distribution  $p_{data}(\mathbf{x})$  to generate novel samples (e.g., text, images, or audio) through an estimated model distribution  $p_{model}(\mathbf{x})$  [5]. Modern AI models are considered Foundational Models, which represent a class of general-purpose models capable of handling diverse data modalities and supporting multiple AI tasks [7]. Notable examples include the Llama 3 family of models, which are foundational language models featuring multilingual capabilities, programming proficiency, reasoning skills, and tool integration [7]. Equation (1) represents the goal of GAI, which is to learn a model distribution  $p_{model}(\mathbf{x})$  that approximates the true data distribution  $p_{data}(\mathbf{x})$ , where  $\mathbf{x}$  represents observations from the data domain.

$$p_{model}(\mathbf{x}) \approx p_{data}(\mathbf{x}) \quad (1)$$

LLMs are a specialized subset of foundational AI and GAI models, with a primary focus on text-based tasks. These models leverage transformer architectures

[12,7] and undergo extensive pre-training on textual corpora. Notably, LLMs exhibit exceptional capabilities in natural language understanding and complex problem-solving (i.e., General Purpose Task Solver) [18], rendering them particularly effective for sophisticated linguistic tasks. The basic working of an LLMs can be thought of as a next-word prediction problem, where given a sequence of tokens  $\mathbf{x}_{1:t} = (x_1, \dots, x_t)$ , an LLMs defines a conditional probability distribution over the vocabulary  $\mathcal{V}$ :

$$P(x_{t+1}|\mathbf{x}_{1:t};\theta) = \text{softmax}(\mathbf{W} \cdot \mathbf{h}_t + \mathbf{b}) \quad (2)$$

$\theta$  denotes the model parameters, and  $\mathbf{h}_t$  is obtained by several layers of transformer blocks that process an embedded representation of the tokenized text:

$$\mathbf{h}_t = \text{Transformer}(\mathbf{x}_{1:t}; \theta) \quad (3)$$

## 2.2 Related Works

There is significant interest within both the scientific community and legal institutions in studying the applications of AI in this expansive field. The current state of the art in NLP with DL primarily focuses on the use of transformer-based architectures, which have demonstrated superior performance across various NLP tasks.

In [17], the deployment of a RoBERTa Base QA classification model specifically for legal questions is discussed. It compares the performance of this transformer model against a traditional linear Support Vector Machine (SVM) in the legal domain, using the PRIVACYQA dataset. The results indicate that RoBERTa significantly outperforms the SVM, achieving a 31% improvement in F1-score and a 41% improvement in Mean Reciprocal Rank, demonstrating the effectiveness of transformers in enhancing answer retrieval for legal inquiries. In [16], the researchers focus on a Transformer-based neural network for answer selection in general QA systems, not specifically in the legal domain. While Transformers are effective in extracting global information and improving answer selection, the research does not address their application in law-related QA. For insights on using Transformers in legal contexts, further research specific to LNLP applications would be necessary. For information retrieval, [9] discuss the use of transformer-based approaches for legal information retrieval and entailment, specifically in the context of the COLIEE competition. For case law retrieval (Task 1), a sentence-transformer model was employed to create numeric representations of case paragraphs, facilitating similarity assessments. In entailment (Task 4), a fine-tuned DeBERTa large language model was utilized, achieving third place among eight teams. These methods demonstrate the effectiveness of transformers in addressing LQA challenges. Another area of interest is predicting legal judgments. In [6] six transformer-based models (BERT, XLNet, RoBERTa, DeBERTa, ELECTRA, and BigBird) are evaluated in predicting legal judgments using the ILDC dataset. It highlights the potential of

these models in automating legal decision-making, with newer models achieving up to 80% accuracy compared to 76% for older models. This comparative analysis underscores the advancements in NLP technology and its applicability in the legal domain, suggesting a promising future for QA in law through transformer models. Finally, in [14] the use of transformers to generate legal text is addressed, indicating their potential to assist in legal practice. The findings suggest that these models can effectively generate judicial opinions and help differentiate between human and machine-generated legal texts, which could be beneficial for tasks like drafting contracts and briefs, although direct application to QA is not discussed.

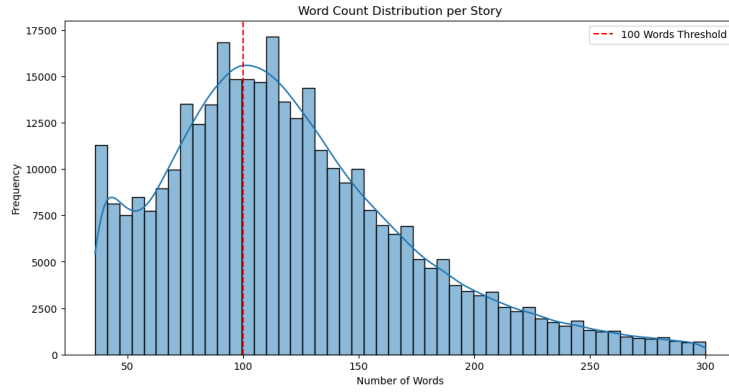
In this section, we briefly discuss foundational concepts and present research related to our work. It is noteworthy that this study introduces a novel application of transformer-based models. To the best of our knowledge, there is currently no evidence of similar studies that simultaneously address both Spanish and legal texts.

### 3 Methodology

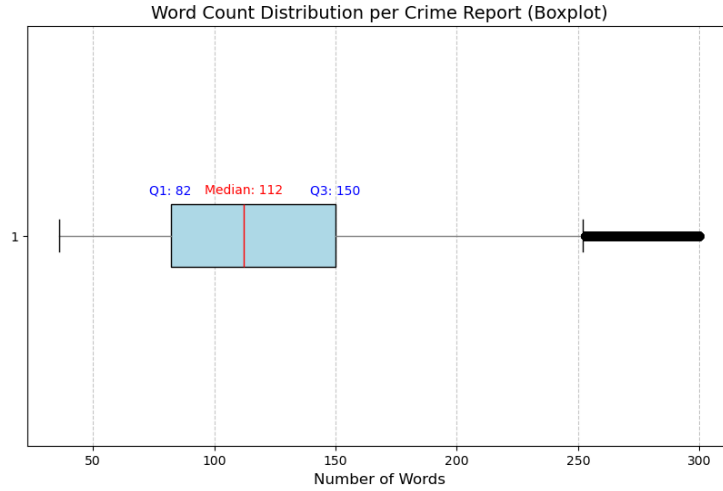
In this paper, we present a BERT-based model fine-tuned for EQA on Spanish-language criminal reports. We started from a Spanish BERT architecture already pre-trained and optimized for QA tasks, then fine-tuned it to extract information from robbery crime narratives. Our methodology consists of three phases: (1) dataset generation, (2) pre-trained baseline model performance, and (3) model fine-tuning.

#### 3.1 Phase 1: Robbery Crime QA Dataset Generation

**Data Collection and Preprocessing** We constructed our dataset using a set of unlabeled robbery crime reports. The text preprocessing pipeline involved removing non-alphanumeric characters and converting all text to lowercase. Statistical analysis of the collected reports revealed an average length (mean) of 112 words, and an upper bound of 150 words. Consequently, we included reports  $x_i$  satisfying  $82 \leq \Gamma(x_i) \leq 150$ , where  $\Gamma : x_i \rightarrow \mathbb{N}$  denotes the word count function for report  $x_i$ . This filtering criterion effectively removed statistical outliers while maintaining the core distribution. Figure 1 illustrates the word count distribution across the original dataset of 174,594 samples, presented through both histogram and box plot visualizations. Finally, we transformed this dataset into a QA format compatible with SQuAD using GAI.



(a) Original Dataset Histogram



(b) Box plot of Words Length in Original Dataset

Fig. 1: Distribution of word counts in collected data visualized through (a) a histogram and (b) a box plot

**Generative AI Prompt Engineering** For this research, we used Meta-LLAMA-3-8B-Instruct[1], an instruction-tuned variant of Meta’s LLAMA 3 8B parameter language model, optimized for conversational and task-oriented interactions with multilingual capabilities. Built on a decoder-only TRANSFORMER architecture, it leverages supervised fine-tuning (SFT) and reinforcement learning from human feedback (RLHF) to align responses with user intent. The model is pre-trained on a diverse corpus of publicly available text data, with enhancements in reasoning, coding, and safety mitigation compared to its predecessors.

Prompt engineering is the process of crafting a suitable prompt to drive the LLMs to solve a complex task, using natural language text [18]. In this work, the LLM functions as an expert data science assistant with the goal of constructing a dataset in the SQuAD format in Spanish. For this purpose, the prompt receives the content of the crime report as the `context` and each of the questions from Table 1 as the `question`. The LLM was also instructed to retrieve a `json` format output with the variables: `context`, `question`, `answer_text`, `answer_start`, `answer_end`, and `impossible_find_answer`. The latter was calculated by finding the `answer_text` in the context. In the case that no answer was possible to be retrieved, either because there is no answer or because the LLM failed, the variable `impossible_find_answer` was set to `True`.

Table 1: Proposed Questions for the Dataset

| Question                                                                 |
|--------------------------------------------------------------------------|
| What objects were stolen?                                                |
| On what date did the incident occur?                                     |
| At what time did the robbery happen?                                     |
| At what address or between which streets did the robbery/incident occur? |
| What was the dollar value of the stolen or taken objects?                |

Hugging Face’s Inference API processed samples at a speed of 6.4 samples per minute. A total of 1,544 samples were obtained after four hours. Our evaluation revealed two key limitations of the LLM: (1) incomplete retrieval of answers for certain queries, and (2) sporadic generation of irrelevant content. These results suggest that manual verification of the generated dataset can aid in improving quality. However, the engineered prompt allowed for the identification of problems in generated samples.

**Dataset Quality** After collecting the initial 1,544 samples, which generated 7,705 question-answer pairs at five questions per context, we conducted quality assessment using the `impossible_find_answer` flag. This process identified approximately 3,000 invalid cases, leading us to retain only samples where `impossible_find_answer` is `False`. Through histogram and box plot analysis of answer lengths, we established an upper limit of 17 words for valid responses. The final curated dataset consisted of 4,572 samples, all containing at least one word and meeting our validity criteria, suitable for model training and evaluation.

**Dataset Train and Test Split** The dataset was partitioned into training and testing sets, with 20% of the samples allocated for evaluation. The remaining 80% were used for training. This approach is a fundamental machine learning technique that helps to evaluate the model’s performance and generalization capability. Table 2 summarizes the sample sizes for each subset. The split was performed randomly.

Table 2: Dataset Train and Test Split

| Partition | # Samples |
|-----------|-----------|
| Train     | 3,657     |
| Test      | 915       |
| Total     | 4,572     |

### 3.2 Phase 2: Baseline Model Architecture and Performance

We selected the `mrm8488/bert-base-spanish-wwm-cased-finetuned-spa-squad2-es` model for fine-tuning due to its specialization for Spanish language processing and QA tasks. This model represents a distilled version of BETO (Spanish BERT) [3] that was further fine-tuned on the SQuAD-es-v2.0 dataset. The architecture follows the standard BERT-base [4] configuration, featuring:

- 12 Transformer blocks
- 768-dimensional hidden states
- 12 self-attention heads
- 384 tokens of sequence length

BETO was pretrained on an extensive Spanish corpus using Whole Word Masking. In preliminary testing without additional fine-tuning, the model achieved an F1-score of 81.70, demonstrating strong baseline performance for Spanish QA tasks.

### 3.3 Phase 3: Model Fine-Tuning

To fine-tune the model, both the question and the answer need to be tokenized. As a result of this process, the indices indicating the start and end of the answer must be recalculated based on the tokenized embedding. In the case of long text, truncation should be applied only to the context, and a window of 128 tokens was used to avoid losing the answer during truncation. However, in this scenario, since most of the current samples have no more than 100 words in the context, this issue is unlikely to occur; nonetheless, it has been addressed in the program.

In order to train the model, the parameters shown in Table 3 were used. A small learning rate is recommended for fine-tuning in order to reuse the pre-training from the original model and adapt the new model to the target dataset. Our results prove that it is possible to achieve a better performance on the target domain with the fine-tuned model rather than using the pre-trained model. Although the difference in performance is small, it could be related to the fact that we were not able to use the whole dataset.

## 4 Results

In this section, we present the results of fine-tuning the model, along with the performance of the pre-trained model on the test set. This comparison was conducted to determine whether the fine-tuned model demonstrated any improvements over the original pre-trained model. Table 4 shows the performance of

Table 3: Training Parameters

| Parameter                       | Value              |
|---------------------------------|--------------------|
| Strategy                        | epoch              |
| Learning rate                   | $2 \times 10^{-5}$ |
| Batch Size                      | 64                 |
| Epochs                          | 10                 |
| Weight Decay                    | 0.01               |
| FP16                            | True               |
| Metric for best model F1 (eval) |                    |

the pre-trained model, while Table 5 shows the performance of the fine-tuned model. It is worth noting that the latter improves upon the former. The fine-tuned model achieved an F1 score of 82.88 on the evaluation dataset. In this case, we use `squad_v2` which contains both the F1-SCORE and EXACT MATCH. A function was prepared in order to be used by the Trainer Class.

Table 4: Pre-trained Model Performance

| Metric                      | Value    |
|-----------------------------|----------|
| eval_loss                   | 0.9102   |
| eval_model_preparation_time | 0.0077   |
| eval_exact                  | 53.5519  |
| eval_f1                     | 81.7041  |
| eval_total                  | 915.0000 |
| eval_HasAns_exact           | 53.5519  |
| eval_HasAns_f1              | 81.7041  |
| eval_HasAns_total           | 915.0000 |
| eval_best_exact             | 53.5519  |
| eval_best_exact_thresh      | 0.0000   |
| eval_best_f1                | 81.7041  |
| eval_best_f1_thresh         | 0.0000   |
| eval_runtime                | 12.3982  |
| eval_samples_per_second     | 73.8010  |
| eval_steps_per_second       | 1.2100   |

## 5 Conclusions and Future Works

In this study, we demonstrated the effectiveness of transformer-based models for EQA within the context of criminal report analysis. Our fine-tuned model showed a notable improvement, achieving an F1 score of 82.88 compared to the pre-trained model’s score of 81.70. This indicates the potential for enhanced performance through domain-specific fine-tuning.

The application of our model to robbery crime reports could facilitate the extraction of crucial information, supporting legal practitioners in their investigations and reporting. However, the study also highlighted some limitations, including the model’s occasional inability to retrieve accurate answers and the need for manual verification of results.

Table 5: Fine-Tuned Model Performance

| Metric                      | Value    |
|-----------------------------|----------|
| eval_loss                   | 1.0088   |
| eval_model_preparation_time | 0.0077   |
| eval_exact                  | 55.7377  |
| eval_f1                     | 82.8805  |
| eval_total                  | 915.0000 |
| eval_HasAns_exact           | 55.7377  |
| eval_HasAns_f1              | 82.8805  |
| eval_HasAns_total           | 915.0000 |
| eval_best_exact             | 55.7377  |
| eval_best_exact_thresh      | 0.0000   |
| eval_best_f1                | 82.8805  |
| eval_best_f1_thresh         | 0.0000   |
| eval_runtime                | 12.8109  |
| eval_samples_per_second     | 71.4240  |
| eval_steps_per_second       | 1.1710   |
| epoch                       | 10.0000  |

Future work will aim to address these limitations by exploring additional advanced techniques and incorporating manual labeling of the dataset by expert peers. Additional computational resources will also be required to process larger datasets. Furthermore, the integration of AI into the legal domain holds significant promise for more efficient case management and resource allocation, ultimately contributing to a safer community. It is also of interest to investigate other NLP techniques within the Ecuadorian legal context, such as Legal Named Entity Recognition (NER) and Legal Document Summarization (LDS). We encourage further research and collaboration to refine these approaches and harness the full potential of AI in supporting the justice system.

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